



Waiting time during admission procedures increases social inequalities in higher education

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Affiliations are included on p. 9.

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Many domains in life require people to wait to access better outcomes, such as waiting in line to access prized tickets for a show, waiting to obtain a job offer from a selective firm, or being waitlisted for a prestigious university. Although waiting might be perceived as a fair mechanism to allocate resources, it may have unintended consequences on social inequalities as individuals from lower socioeconomic backgrounds (SES) may be less likely to wait. In this study, we investigate socioeconomic differences in waiting behavior during the higher education admission process in France. To do so, we leveraged a unique national database that recorded the admission behavior of every high schooler in France over a period of three years. During the admissions process, students faced a trade-off: They could either accept an immediate offer from a university or wait in the hope of receiving a preferred offer later. Results collected among 274,316 students reveal that low-SES students were significantly less likely to wait for preferred offers, even when controlling for objective costs and benefits of waiting. This led low-SES students to ultimately enroll in programs they liked less and that were less prestigious. These results highlight how admission procedures generating uncertainty and requiring students to wait may inadvertently exacerbate educational inequalities. Such systems, while efficient, can end up favoring students who wait more. This research contributes to current discussions on admission procedure fairness by illustrating that seemingly neutral algorithms can in fact deepen social inequalities if their design rewards patience.

waiting | social inequalities | time discounting

Many life outcomes depend on people's ability to wait. For example, spots in prestigious universities may be gradually given out to waitlisted students. Having a good job may require people to wait several months before obtaining the right offer. Having access to specialized healthcare may require waiting patiently to get an appointment. Enrolling children in a nearby daycare may require parents to endure long waiting times. These situations present a trade-off between accepting an offer now (a spot in a university, a job offer, a doctor's appointment, or a daycare spot) or waiting to obtain something better (a spot in a more prestigious university, a higher paying job, an appointment with a specialist, a place at a closer daycare). In many situations, waiting is beneficial, it improves people's outcomes (1–3) and can have a large effect on people's lives because of compounding effects. For example, the selectivity of the university people attended can have a long-lasting effect on students' career and earnings (4, 5), and people's first salary typically has a long-lasting effect on their lifelong earnings, a phenomenon referred to as “wage scarring” (6).

Many public goods such as plots in community gardens, spots in public preschools and daycares, access to rent-controlled apartments, and even citizenship applications are allocated through waitlists. Although these procedures may seem more equal than allocating goods through financial mechanisms, they might have negative distributive consequences. If people's ability to wait is influenced by their socioeconomic status, then systems that use waitlists to allocate goods may inadvertently deepen social inequalities.

A large literature has shown that people from lower socioeconomic backgrounds tend to be less likely to wait for later and larger rewards compared to people from higher socioeconomic backgrounds in experimental contexts. A recent meta-analysis of 248 effects from 70 independent studies involving over 35,000 participants confirmed that individuals raised in adverse environments tend to favor immediate rewards in laboratory experiments (7). These studies rely on behavioral tasks to investigate people's ability to wait, where participants are typically asked to choose between smaller rewards available immediately (e.g., receiving 10 euros today) and larger rewards delayed in time (e.g., receiving 20 euros, but in a week) (8–11). Studies have also identified a causal relation between people's socioeconomic status and their ability to wait in laboratory settings where perceived

Significance

Do resource allocation systems that reward waiting unintentionally deepen social inequalities? Getting access to resources like university admissions often requires people to wait. Yet, the impact of social class on waiting to access these resources is understudied. Most existing studies on waiting behavior focus on lab experiments with money-based rewards. In this study, we examine real-world behavior in the higher education admission process. We find that students from lower-income backgrounds are less likely to wait for better opportunities, even when waiting could lead to admission to more prestigious schools. This shows that systems that require waiting can unintentionally make social inequalities worse. To promote fairness, we recommend designing allocation systems that reduce waiting times or encourage students to wait.

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socioeconomic status is experimentally manipulated (12–14), using natural experiments such as natural catastrophes affecting people's income (15–17), or field experiments involving cash transfers (18–20).

These socioeconomic differences in waiting behaviors can be attributed to different costs and benefits associated with waiting, as well as psycho-social factors. Differences in waiting behavior may reflect socioeconomic differences in time discounting (8, 21, 22), that is, how much people value future outcomes compared to present outcomes. Individuals who value the future less are less likely to wait for better outcomes. Differences in waiting behavior may also stem from differences in how much people trust the future, for example, if they have low self-confidence, or if they do not trust others, they may not believe that waiting is worthwhile (23, 24). People's willingness to wait may also depend on their uncertainty aversion (25), as people may not want to delay reaching a decision. Finally, people's waiting behavior may be influenced by their understanding of the mechanism. People who have access to less information may not realize that waiting is optimal.

The existing literature therefore suggests that resource allocation systems that reward waiting may unintentionally deepen social inequalities. The purpose of this study is to investigate waiting behaviors in the context of higher education admissions in France. Investigating socioeconomic differences in waiting behavior in real-life settings can be tricky as individuals' socioeconomic status may influence many aspects of the waiting decision. Individuals from lower socioeconomic backgrounds may be less likely to wait because their likelihood of obtaining a later reward is or seems lower, the value of the delayed reward is or seems lower, or waiting comes at a financial cost. For example, individuals from a lower socioeconomic background may decide to accept the first job offer they receive because they are less likely to receive a later job offer, prospective jobs all offer the same salary, or they cannot afford to delay their first pay check. Understanding socioeconomic differences in waiting behaviors beyond these objective costs and benefits requires keeping these constant across individuals.

This French higher education admission process offers a unique opportunity to measure socioeconomic differences in waiting behaviors as students face a choice: They can either accept the first offer they receive from a higher education program, or they can wait a few weeks in the hope of receiving an offer from a program they prefer. Specifically, students who are waitlisted in programs that they rank higher in their list of wishes can act in one of three ways: They can accept the first offer they receive, they can wait

two weeks until the second admission phase and accept the offer they receive then, or they can wait an additional 2 wk until the third admission phase and accept the final offer they receive, as shown in Fig. 1. There are no direct financial costs for waiting and no risk for students, who are guaranteed a spot in the program from which they receive their initial offer. As a result, given that waiting a few weeks can only lead students to be accepted in programs they prefer, the rational decision would be to always wait.

This study examines whether differences in students' waiting behavior deepen social inequalities. To answer this question, we retrieved data from the centralized admission platform (called Admission Post Bac, or "APB") on which all high school students applied to higher education programs, for a total of about 400,000 students per year. In line with rational choice models (26), we consider that the decision to wait may be influenced by rational factors such as the probability of getting the reward, the value of the reward, and the cost of waiting. This database allowed us to control for such factors as it includes information about students' academic performance (probability of getting the reward) and the prestige and preference gain for being accepted in a pending program (value of the reward). There are no direct waiting costs in the admission system. However, because cheap student housing may run out quickly, students who wait a few weeks before enrolling in a program and searching for housing may pay a higher price. Therefore, students who wait a few weeks may face additional financial costs. Therefore, we consider that waiting may lead to higher costs if it requires the student to delay finding housing (cost of waiting). To measure a student's socioeconomic status, we rely on the APB database which contains information about the student's financial aid status. In France, high-school financial aid is awarded to all families belonging roughly to the 20th lowest income percentile, regardless of students' academic abilities. In our analysis, we used financial aid status as a proxy for having a low socioeconomic background, with 18.3% of students in our final sample receiving financial aid.

We hypothesized that students from lower socioeconomic background are less likely to wait to be accepted into a preferred program. We first showed that the probability of being accepted into a higher education program was independent from students' socioeconomic status, once academic performance and the selectivity of the program had been accounted for. We then hypothesized that after controlling for rational choice factors, socioeconomic differences in waiting behavior persist, which contributes to education inequalities.

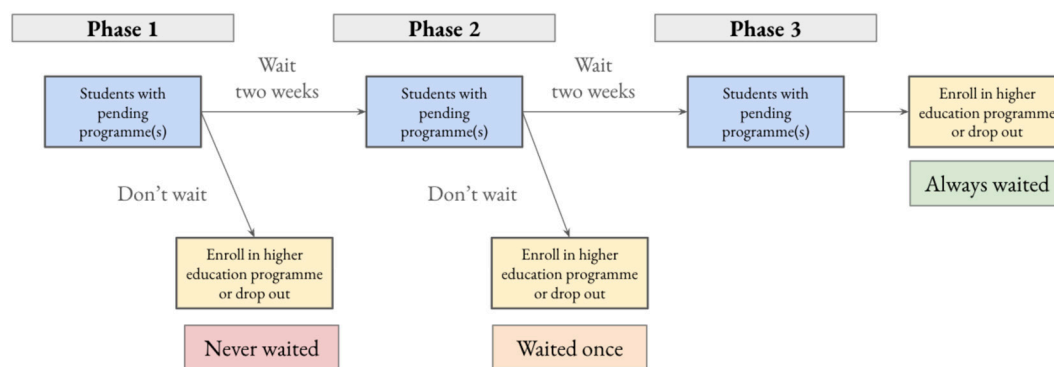


Fig. 1. The French centralized admission process for higher education. The figure represents the three phases of the higher education admission process, and students' possible decisions to wait or not at each phase. The process was as follows. Students submitted a rank-ordered list of wishes in May, prior to Phase 1, which took place early in June. During Phase 1, all students received an initial offer from a higher education program. Some students were also waitlisted for other programs, meaning that they had pending wishes. These students could decide between accepting the initial offer and enrolling in the program or waiting until the next phase (2 wk later) to see whether they would be admitted into a pending program. The same process took place in Phase 2. Finally, during Phase 3, students could either accept an offer or drop out of the admission system. Depending on their decisions in each Phase, students can be categorized as having "Never waited," "Waited once," or "Always waited."

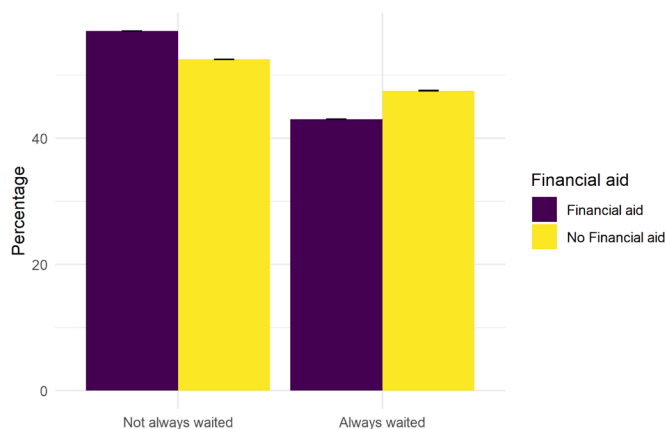


Fig. 2. Waiting behavior by financial aid status. Waiting behavior according to financial aid status for the 2015, 2016, and 2017 cohorts. A greater share of financial aid beneficiaries did not always wait (either didn't wait or waited once), while a greater share of non-financial aid beneficiaries always waited. SE are shown in the black bars. All differences are significant.

Results

Our analysis included 274,316 students who were waitlisted for at least one of their wishes. Among those, we found that about 23.2% of students never waited, 30.1% of students waited only once, and 46.7% of students always waited. We first investigated socioeconomic differences in waiting behavior. To do so, we grouped students into students who did not always wait (accepted an offer during Phase 1 or 2) and students who always waited (accepted an offer during Phase 3). We found that consistent with our hypothesis, financial aid beneficiaries were 5 percentage points less likely to wait to be accepted into a preferred program (SE = 0.002, $P < 0.001$), see Fig. 2.

We then investigated whether socioeconomic status affected the likelihood of being admitted into a pending program. If this is the case, socioeconomic differences in waiting would simply reflect the fact that some students are less likely to be admitted into a pending program, such that waiting is not worthwhile. In France, higher education programs have to rank all students who are waitlisted. We considered students' ranks in each program they applied to and combined this with the admission cutoff at the end of the admission process to estimate their likelihood of admission. This means that for each student, we could estimate whether they would have been admitted into a program, regardless of their waiting behavior. Because academic performance and the prestige of the pending program may affect the chances of admission, we controlled for these factors. To measure academic performance, we used the Baccalaureate grades, a centralized final exam that all students take at the end of high school.

We found that there was no significant difference between financial aid beneficiaries and the rest of the students in terms of their likelihood of being admitted into a pending program, whether during Phase 2 or during Phase 3, see Table 1. This means that socioeconomic status did not affect the likelihood of being admitted into a pending program, and differences in waiting behavior cannot be explained by students' correct anticipation of not being admitted. Said differently, had they waited, students from lower socioeconomic backgrounds would have been as likely as other students to be admitted into pending programs.

We confirmed this analysis by looking at how students' waiting behavior affected their likelihood of being admitted into preferred programs, according to their socioeconomic status. We found that students who waited once increased the rank of their final offer by an average of 0.64 (SE = 0.01, $P < 0.001$) while those who

always waited increased the rank of their final offer by an average of 0.51 (SE = 0.005, $P < 0.001$), a large increase considering that on average students applied to 11 programs, see Table 2. We found that the increase in the rank of the final offer was not significantly different between financial aid (FA) and non-financial aid beneficiaries. This means that waiting led to comparable gains in terms of better-ranked offers for all students.

We then investigated whether socioeconomic differences in waiting depended on the actual costs and benefits of waiting, by sequentially adding variables to the analysis. First, we tested the role of the benefit of waiting, which derives from the probability of being admitted to a pending program and the value of this program. The rationale was that students would get a larger benefit from waiting if their chance of being admitted to a pending program was higher and/or if the value of this pending program was higher. We estimated students' chances of being admitted by their high-school track and their grades at the Baccalaureate. In France, students must choose between three types of high-school tracks that provide different curricula and prepare for different opportunities: the "General track" prepares for university education, the "Technological track" prepares for short higher-education programs, and the "Professional track" prepares students for a vocational career. The general track is further divided into a "Science track," an "Economic and social science track," and a "Humanities track." Students can apply to higher education programs regardless of the high school track they attended. Nevertheless, tracks are highly hierarchical, and national tests show that students' average academic abilities differ in the various tracks. Hence, the level of preparedness and likelihood of being accepted vary consistently depending on the track. We found that financial aid beneficiaries were 5 percentage points less likely to always wait (SE = 0.002, $P < 0.001$) even after accounting for differences in academic performance, see Table 3. This means that even when they had comparable chances of being admitted to a pending program, financial aid beneficiaries waited less than their peers.

Table 1. Chances of admission into a pending program according to academic performance, prestige of the program, and financial aid status

Chances of admission into pending program		
	Dependent variable:	
	Phase 2	Phase 3
	(1)	(2)
Bac grade	0.13*** (0.01)	−0.11*** (0.01)
Prestige of pending program	−0.08*** (0.02)	−0.06* (0.02)
Financial aid beneficiaries	0.01 (0.05)	−0.03 (0.06)
Year fixed effect	Yes	Yes
Observations	274,316	203,600
R ²	0.002	0.002
Adjusted R ²	0.002	0.002
Note:	* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$	

We regressed student's chances of being admitted into a pending program during Phases 2 and 3 according to their academic performance, the prestige of the pending program, and their financial aid status. We found that financial aid status did not influence chances of admission, suggesting that waiting is worthwhile for all students. We included year fixed effects with 2015 as the baseline. SE were clustered at the high school level. *5%, **1%, ***0.1% significance level.

Table 2. Effect of waiting on the rank of the program where students enroll

Effect of waiting on the rank of enrollment

	Dependent variable:
	Preference gain
Waited once	0.64*** (0.01)
Always waited	0.51*** (0.005)
Waited once * FA	0.01 (0.02)
Always waited * FA	-0.01 (0.01)
Mean (Never waited)	0
Included FA beneficiaries dummy	Yes
Year fixed effect	Yes
Observations	274,316
Adjusted R ²	0.03

We regressed students' gain in terms of the ranking of the program in which they end up enrolling as a function of the students' waiting behavior (Never waited is the reference, Waited once, and Always waited are dummies) and financial aid status. Waiting led students to enroll in higher-ranked programs, and there was no significant difference in the gain in terms of rank between financial aid beneficiaries (FA) and non-financial aid beneficiaries. We included year fixed effects with 2015 as the baseline. SE were clustered at the high school level. *5%, **1%, ***0.1% significance level.

We then investigated whether differences in waiting behavior remained after taking into account the value of the pending programs in addition to the probability of being admitted. The rationale was that students who stand to gain more from being admitted to a pending program would be more motivated to wait. We measured the value of waiting as the number of pending programs (having more pending programs increases the value of waiting), the ranking of the student on the waitlist of the pending program (students ranked higher stand to have higher benefits from waiting), how student ranked the offer they received during Phase 1 (students who received initial offers they did not rank very high stand to gain more from waiting for another offer), the prestige of the offer students received during Phase 1 (the more prestigious the initial offer, the lower the value of waiting), and the gain in prestige from being accepted in a pending program (the greater the gains, the greater the value of waiting). We found that financial aid beneficiaries were 6 percentage points less likely to always wait (SE = 0.002, $P < 0.001$) even after accounting for differences in the value of waiting in addition to the probability of being admitted into a pending program, see Table 3.

Second, we examined whether students' behavior was affected by the cost of waiting. There is indeed no direct financial cost of waiting, but a higher distance between the initial offer received by the students and the pending program may lead to additional housing costs. The rationale behind this is that in France, spots in affordable student housing are scarce, and may all be allocated within a few weeks. Students who enroll later in a program may no longer find affordable housing next to that program. As a result, delaying the decision to enroll in a program may come at a financial cost. Students who are waiting to hear back from programs that are located next to their initial offer can already look for housing, but students who are waiting for programs located far away may have to wait before finding housing. To take this into account, we regressed the probability of waiting on the distance between the initial offer received by the student and the pending program. We predicted that when the distance is greater, it decreases the chances of finding affordable housing.

Table 3. Socioeconomic differences in waiting behavior, taking into the costs and benefits of waiting

Socioeconomic differences in waiting, controlling for waiting costs and benefits

	Always waited		
	(1)	(2)	(3)
Financial aid beneficiaries	-0.05*** (0.002)	-0.06*** (0.002)	-0.05*** (0.003)
Technological track	-0.06*** (0.003)	-0.06*** (0.003)	-0.06*** (0.003)
Professional track	-0.08*** (0.003)	-0.11*** (0.004)	-0.11*** (0.004)
Science track	-0.04*** (0.003)	0.01*** (0.003)	0.01*** (0.003)
Humanities track	-0.06*** (0.004)	-0.06*** (0.004)	-0.06*** (0.004)
Baccalaureate grade	-0.02*** (0.0004)	0.001** (0.001)	0.001* (0.001)
Number of pending programs		0.03*** (0.001)	0.03*** (0.001)
Chances of being accepted in a pending program		-0.001*** (0.0001)	-0.001*** (0.0001)
Rank offer phase 1		0.004*** (0.0003)	0.004*** (0.0003)
Prestige offer phase 1		-0.03*** (0.001)	-0.03*** (0.001)
Gain in prestige from pending program		-0.003* (0.001)	-0.003 (0.001)
Distance between offer and pending (100 km)			-0.001*** (0.0002)
Distance between offer and pending (100 km) * FA			-0.0002 (0.0003)
Mean nonfinancial aid beneficiaries	0.48	0.48	0.48
Year fixed effect	Yes	Yes	Yes
Observations	274,316	274,316	258,832
Adjusted R ²	0.06	0.08	0.08

This table shows the results of linear regressions of a dummy indicating whether students decided to always wait (1) instead of never waiting or waiting once (0). The independent variables are a dummy indicating whether students received financial aid during high school, dummies for students' high-school track (the reference is the Economic and Social Science track), the Baccalaureate grade, the number of pending programs, the chances of being accepted in a pending program, the rank of the offer during Phase 1, the prestige of the offer during Phase 1, the gain in prestige from being accepted in a pending program, the distance between the offer received during Phase 1 and the pending program, and an interaction term between this distance and students' financial aid status of (FA). Financial aid beneficiaries were less likely to decide to always wait, even when taking into account the costs and benefits of waiting. We included year fixed effects with 2015 as the reference. SE were clustered at the high-school level. *5%, **1%, ***0.1% significance level.

Table 4. Socioeconomic differences in waiting behavior, controlling for academic performance, aspirations, waiting costs, and high school fixed effects

	Always waited	
	(1)	(2)
Financial aid beneficiaries	−0.05*** (0.003)	−0.04*** (0.003)
Mean non-financial aid beneficiaries	0.48	0.48
Control for academic performance	Yes	Yes
Control for aspirations	Yes	Yes
Control for waiting cost	Yes	Yes
Year fixed effect	Yes	Yes
High school fixed effect	No	Yes
Observations	258,832	258,832
Adjusted R ²	0.08	0.10

This table shows the results of a linear regression of a dummy indicating whether students decided to always wait (1) instead of never wait or wait once (0). The independent variables are all variables linked to academic performance, aspirations, and waiting costs (hidden), a dummy indicating whether students received financial aid during high school, plus dummies indicating the high school of the student (hidden). Financial aid beneficiaries were 4 percentage points less likely to decide to always wait, even after controlling for academic performance, aspirations, waiting costs, and unobservable factors linked to students' high school such as peer effects, neighborhood effects, or teacher effects. We included year fixed effects with 2015 as the reference. SE are clustered at the high school level. *5%, **1%, ***0.1% significance level.

This should make all students, and especially financial aid beneficiaries, less likely to wait. Given that some higher education programs offer free housing to admitted students, we considered waiting for admission into these programs as having no costs. Consistent with our hypothesis, we found that students were less likely to wait when their pending program was far from their initial offer. However, this effect was not stronger among financial aid beneficiaries. After taking waiting costs into account, financial aid beneficiaries were still 5 percentage points less likely to always wait ($SE = 0.003$, $P < 0.001$), see Table 3 for results. This shows that socioeconomic differences in waiting behavior persist even after taking into account rational choice factors captured by the differences in the probability of being admitted, the value of the pending programs and waiting costs. This suggests that other psycho-social factors may account for socioeconomic differences in waiting behavior.

We wanted to isolate the effect of a student's school environment and test whether socioeconomic differences in waiting persisted even between students from the same high school. High school effects include peer influences, where a student's enrollment decision depends on their peers' choices, or informational barriers, where students from more affluent schools receive better advice on how to navigate pending offers. Our analysis showed that socioeconomic differences in waiting remained even when including high school fixed effects in addition to rational choice factors, with financial aid beneficiaries being 4 percentage points less likely to always wait ($SE = 0.003$, $P < 0.001$), see Table 4, suggesting that other psycho-social factors are at play.

To ensure the robustness of our results, we examined whether altering the dependent variable affected the findings by reclassifying waiting behaviors. Specifically, we compared students who had never waited to those who had waited at least once and found similar results (*SI Appendix, section 2*). We found that socioeconomic differences in waiting behavior persisted even when reclassifying waiting behaviors.

Because socioeconomic status is a complex construct that may not be entirely captured by financial aid status, we also replicated our results using parental occupational category as a proxy for socioeconomic status. Specifically, we classified students into four groups based on their parents' occupations and analyzed the effect of having parents in the lowest occupational category. We found similar socioeconomic differences in waiting behavior using parental occupational category as an indicator of social class, as detailed in *SI Appendix, section 3*. Specifically, we found that students whose parents belonged to the lowest social class were 3 percentage points less likely to always wait ($SE = 0.003$, $P < 0.001$) after taking the costs and benefits of waiting and high school fixed effects into account.

Discussion

We found that even over short timeframes (2 to 4 wk), students from lower socioeconomic backgrounds were less likely to wait than their peers. This pattern persisted even after controlling for the costs and benefits of waiting for acceptance to a pending program. Our dataset includes the entire student population of France across three academic years, providing high internal validity. In addition, these results are likely applicable to other countries with similar admission systems that require students to wait, such as Ireland, Brazil, the United States, or Sweden.

Our study shows the existence of socioeconomic differences in waiting behavior that cannot be explained by factors such as academic performance, aspirations, financial constraints, or the influence of peers and school environment. These differences in waiting behavior contribute to social inequalities as students from lower socioeconomic backgrounds are less likely to wait for improved academic outcomes. Although previous studies have found an important role of the costs and benefits of higher education choices in explaining socioeconomic differences in behavior (27–30), these factors did not close the waiting gap in our data, suggesting that other psycho-social factors play an important role. Interestingly, when taking into account unobserved characteristics linked to a student's high school, such as neighborhood effects, teacher effects, and peer effects, the waiting gap does not narrow much. This suggests that students' school environment has little influence on their decision to wait.

What could explain these socioeconomic differences in waiting behavior? Several explanations have been proposed in the literature. First, students from lower socioeconomic backgrounds may be less familiar with the admission process and the optimal strategies to adopt. Previous research has shown that students from more affluent backgrounds are more likely to adopt effective strategies when applying to higher education programs (31–33), while those from lower socioeconomic backgrounds often face informational barriers regarding the costs and benefits of higher education (34). Additionally, students from wealthier families typically receive guidance from parents, teachers, or peers on the best strategies to follow (35, 36). Second, students from lower socioeconomic backgrounds may have lower trust in the admission platform due to negative past experiences with public services (37) leading them to secure a spot in a program rather than risk waiting. Research has shown that trust in institutions and administrative processes is often related to socioeconomic status (38–40), which may cause disadvantaged students to adopt a more cautious approach. Third, a large literature has documented lower self-confidence among students from lower socioeconomic backgrounds (41, 42), which could lead them to underestimate their chances of admission to a preferred program or their chances of success in that program (43), thereby reducing the perceived benefits of waiting. Finally, the need to reduce uncertainty may explain some of these differences in waiting behavior (44, 45). Waiting can be seen

as imposing a cognitive tax on students who have to deal with the uncertainty of their higher education pathway. One limitation of our study is that we were unable to differentiate between these different mechanisms. Future research should explore these factors further, for example, by testing whether giving students information about their chances of admission to a program affects their behavior. This could reveal if differences in waiting behavior are linked to self-confidence.

This study contributes to the extensive literature on social inequalities in higher education by identifying a novel factor that deepens inequalities: the behavioral structure of the admission algorithm. Admission algorithms that reward waiting may inadvertently deepen social inequalities as students from lower socioeconomic backgrounds are less likely to wait. These differences in waiting behavior during the admission process can have long-lasting impacts on students' earnings trajectories, as a small increase in the quality of their higher education program affects future earnings among both high and low-socioeconomic students (5, 46–52). For example, a study conducted in Norway found that 23% of variance in earning outcomes can be explained by differences in the quality of the institution in which people had been trained (53), while a study conducted in Chile found that small differences in selectivity were responsible for an increase of average earnings, from 4.7% in the bottom selectivity quartile to 24.2% in the top quartile (54). In the case of the APB platform, the sequential admission procedure allows for more flexibility because students can be waitlisted and offered a spot as other students change their plans. This procedure has been shown to improve outcomes for all students (55). Yet, this added flexibility may also come at a cost. The sequential assignment procedure leads to negative redistributive consequences by favoring students who are more likely to wait.

Reducing inequalities in higher education may thus be achieved through reducing the waiting time in the admission process or encouraging students to wait longer. For example, instead of having a 2-wk period between each admission phase, this waiting time could be shortened to one or two days to limit the impact of waiting. Another solution would be to have waiting as the default option. Alternatively, providing students with information about the importance of waiting may also increase the likelihood of students from lower socioeconomic backgrounds to also wait. Because students from lower socioeconomic backgrounds may have lower self-confidence in their academic abilities, sharing information about their chances of being admitted into a pending program may also increase their likelihood of waiting. Finally, information treatments aimed at improving trust in the admission procedure may also increase the probability of students from lower socioeconomic backgrounds to wait as they no longer fear being left with no offer at all. Future research could investigate the impact of these different options on reducing the socioeconomic gap in waiting.

Our findings also make an important contribution to the literature by providing a real-life test case of socioeconomic differences in waiting behavior. Beyond economic games and laboratory experiments, we found that in a real-life context, students from lower socioeconomic backgrounds were less likely to wait for a better outcome, even though waiting came at no risk and no direct financial costs. Our findings suggest that socioeconomic differences in waiting behavior are due to factors unrelated to a rational choice model or a student's budget constraints. Given the widespread use of waiting as a selection mechanism in real-life situations, waiting times may exacerbate inequalities rather than serve as a cost-free opportunity for low-income individuals to access

services. Our study shows that the capacity to wait often depends on social and psychological resources that individuals from lower socioeconomic backgrounds may lack. In addition, our findings suggest that socioeconomic differences in waiting behavior may be better explained by differences in the perceived rather than objective costs and benefits of waiting. This is consistent with the large literature in behavioral and psychological sciences suggesting that perceived costs and benefits differ according to people's socioeconomic status (56).

Finally, our research contributes to the growing conversation on algorithmic fairness. Algorithms can be biased not simply because of the data they are trained on or the decisions they make, but also because of their very design. If allocation algorithms reward patience, they may accidentally reinforce socioeconomic inequalities, as we show in the case of higher education admissions in France. From an applied point of view, our results provide a rationale for reducing waiting times in higher education admission processes and more generally in all matching markets, such as job markets.

Methods

The French Higher Education System. Between 2009 and 2017, higher education admissions in France were facilitated by a centralized system known as Admission Post-Bac (APB). Students were required to apply through this platform by submitting a list of desired programs, ranked according to their preferences. Over the years, the number of wishes students could rank fluctuated, from 36 in 2015 to 24 in 2016 and 2017. The deadline for submitting the rank-ordered list of wishes was May 31st. While this platform was the main application route for the majority of universities [covering 90% of new high school graduates entering higher education (57)], some private French institutions offered the option to bypass the centralized system. Most higher education programs are highly subsidized by the French state. Admission fees are typically around 180 €/year for the 63% wealthiest students, while they are free for the remaining 37%.

Offers from programs were communicated to students during three phases. The exact timing of each phase varied from 1 y to another but corresponded roughly to the following periods: i) Phase 1: second week of June, ii) Phase 2: fourth week of June, and iii) Phase 3: second week of July. During Phase 1, all students received a single offer from a higher education program. Students who received an offer from their preferred wish (ranked first), could either accept the offer (final yes) or drop out of the admission process entirely (final no). Students who did not receive an offer from their preferred wish faced two possible scenarios. If students had no pending programs, i.e., they were not waitlisted for any other wish, they could either accept the offer (final yes) or drop out of the admission process entirely (final no). If students had at least one pending program, i.e., they were waitlisted on at least one other wish that was ranked higher than the offer they received, they could choose among one of the four following answers: i) Final yes: The candidate accepts the offer, enrolls in the program, and is removed from the waitlists of all their pending wishes. The candidate will thus not receive any other offer in the following phases; ii) Final no: The candidate refuses the offer and all potential future offers; iii) Provisional yes: The candidate accepts the offer but does not enroll in the proposed program. In case the candidate receives a better offer from a pending program in the following phase, they can enroll in that program; iv) Provisional no: The candidate rejects the offer but remains in the system, and in case the student receives an offer from a preferred program, they can enroll in that program.

The choice of provisional yes or provisional no has no direct cost: if the candidate chooses this option, the best-case scenario is that they end up enrolling in a program they had ranked higher in their wish list, and the worst-case scenario is that they enroll in the program that was initially offered. During Phase 2, the same process was repeated for students who had previously given a provisional yes or a provisional no answer. During Phase 3, students could no longer give a provisional answer and had to either accept the offer they received (final yes)

or fully drop out of the platform (final no).^{*} The structure of the admission algorithm allows us to observe student's waiting behavior. We focus our analysis on students who had at least one pending program, as they could decide between waiting to have a better option (provisional yes or provisional no) or not waiting (final yes or final no).

Data and Measures.

Ethics information. The project is a cross-sectional observational study that relies on secondary analyses of data collected by the centralized admission platform APB, a publicly owned and managed platform. The data are protected by a secure bubble technology, managed by the national research data management tool called the CASD, which complies with the GDPR and CNIL security standards and ensures the anonymity of the data. The use of the data for research purposes is regulated by the French Statistical Confidentiality Committee (Comité du Secret Statistique), which approved our project (project scss-3899-1, approval M1569 27/10/2022). Given that the data are secondary, that the CASD oversees data anonymization, and the French Statistical Confidentiality Committee approved our project, no further ethical approval was required given French regulations.

Sample. We conducted a pilot study on the 2015 APB cohort on a sample of 89,263 students who had the opportunity to wait. Among those, 16,165 (18.1%) students received financial aid during high school and 73,098 (81.9%) students did not receive financial aid. Based on this pilot study, we pre-registered the replication of the same analysis on the 2016 and 2017 cohorts. Our pre-registration can be found at this link: <https://osf.io/tcfs7/>.

We then performed an a priori power analysis with the R package pwr. We estimated that for a sample size of 80,000 participants, with 4220 predictors (this includes high-school fixed effects, with 4202 high schools in 2015), an α -level of 5%, a statistical power of 95%, we could detect an effect size of Cohen's $d = 0.004$, based on a two-tailed test. We found in our pilot study an effect of socioeconomic differences in psychological factors on waiting behavior of Cohen's $d = 0.11$, meaning that we were sufficiently powered. Given that the number of students using the APB platform is stable from one year to another, our preregistered study had enough power to detect socioeconomic differences in waiting in the 2016 and 2017 cohorts.

The final sample of students included in our analysis is based on the following criteria: First, we considered all students in their final year of high school (All students). We then selected students who had applied to at least one program on the APB platform and received at least one offer during the first admissions phase. We further restricted our sample to students who had the opportunity to wait, i.e., students who did not get admitted in their top wish and who had at least one pending wish (Target sample). We then excluded all students for whom we had one or more missing values on the variables of interest (financial aid status, high-school track, Baccalaureate grade, number of pending wishes, chances of being accepted in a pending wish, rank of offer received during the first phase, prestige of offer received during the first phase, waiting behavior during each phase) (Final sample). See Fig. 3 for an overview of the funnel. To assess the implications of this sampling strategy, we compared the characteristics of the final sample with the characteristics of the target students and those of all students. See *SI Appendix, Tables SI.1.1 and SI.1.2* for more details on the sample selection and the characteristics of the final sample.

Waiting behavior. We focused our analysis on students who had at least one pending program, meaning that they had the opportunity to wait to receive a preferred offer. We recorded students' waiting behavior during each phase and divided them into three groups: students who never waited (they accepted their offer or left the admission process entirely during Phase 1), students who waited once (they waited during Phase 1 and then accepted their offer or left the admission process entirely during Phase 2), and students who always waited (they waited during Phases 1 and 2), see *SI Appendix, Table SI.1.2* for more details.

^{*}The application process also contained a parallel admission procedure that allowed students to submit new applications to programmes starting from the beginning of phase 2 (called "procédure complémentaire"). For the purpose of our analysis, we do not include the parallel admission procedure for the following reasons. First, students applying in the parallel admission procedure did not have to rank their wishes, so that we cannot know which programmes they preferred. Second, the parallel admission procedure mixes students who are patient enough to apply again to new programmes, and students who regret their initial answer or made a mistake when they answered. We therefore cannot know whether their behavior is the result of more waiting, or of mistakes. Given that we are interested in measuring students' waiting behavior, we do not include the parallel procedure in our analyses.

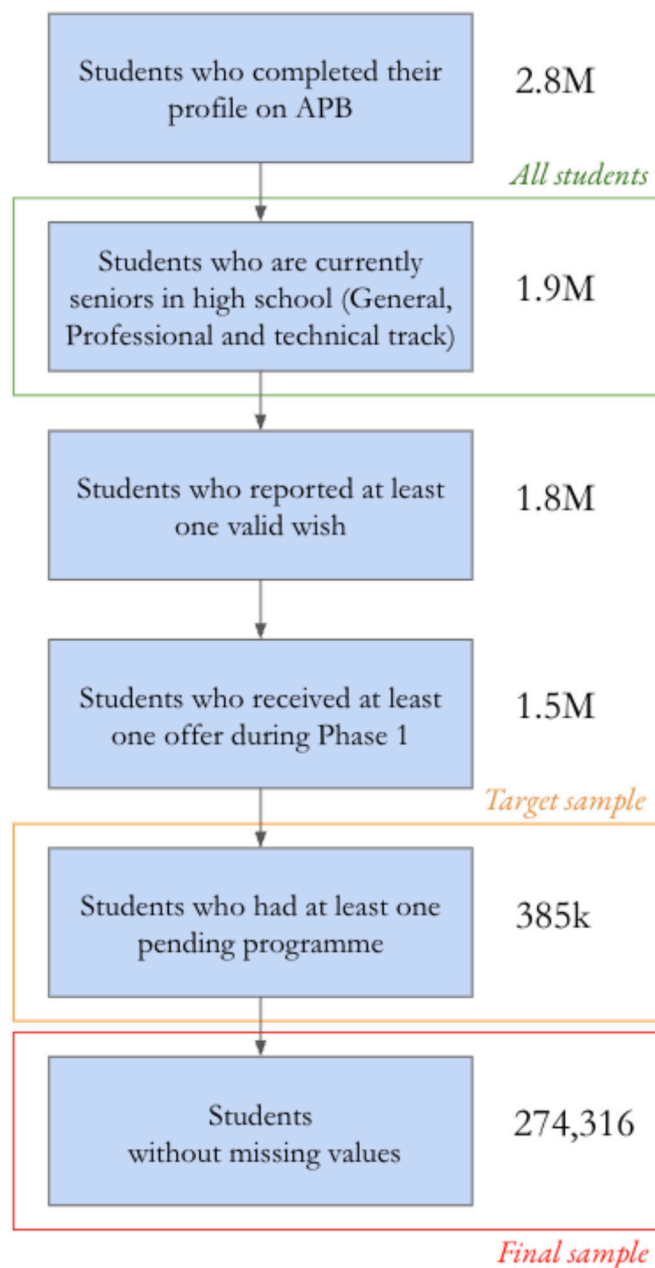


Fig. 3. Sample selection. In 2015, 2016 and 2017, about 2.8 million profiles were created on the centralized admission platform APB. Of those, about 1.9 million were students in their senior year of high school applying to a higher education program (which we later refer to as "All students"). We selected only seniors in high school (General, Professional, and Technical tracks) who reported at least one valid wish and who received an offer during the first phase of admissions, leading to a sample of about 1.5 million students. Because we were interested in studying waiting behaviors, we restricted our sample to students who had the opportunity to wait, that is students who had at least one pending wish. Of this sample of about 385,000 students (which we call our "Target sample"), we further restricted our analysis to students for which we did not have any missing values, leading to a final sample size of 274,316 students.

Socioeconomic status. In France, high school financial aid is awarded to all families belonging roughly to the 20th lowest income percentile, regardless of students' academic abilities. In our analysis, we used financial aid status as a proxy for low socioeconomic background. Close to the national average, 18.3% of students in our final sample received high-school financial aid.

Likelihood of admission into pending program. In France, all higher education programs are required to rank every student who is waitlisted. By combining each student's rank in a given program with the final admission cutoff of the program, we could estimate their chances of admission. This allowed us to determine,

Table 5. Differences in academic performance according to financial aid status

Variable	Mean non-FA	FA - non-FA	P value
Technological track	0.20	0.099	$P < 0.001$
Professional track	0.09	0.110	$P < 0.001$
General track	0.70	-0.209	$P < 0.001$
Economic and social science track	0.23	-0.036	$P < 0.001$
Science track	0.40	-0.182	$P < 0.001$
Humanities track	0.08	0.009	$P < 0.001$
Nonadjusted baccalaureate grade	12.60	-0.976	$P < 0.001$
Baccalaureate grade	12.77	-1.640	$P < 0.001$

This table shows the difference between non-financial aid beneficiaries (non-FA) and financial aid beneficiaries (FA) in terms of academic performance. Financial aid beneficiaries are more likely to be in the technological professional and humanities track compared to non-financial aid beneficiaries. Financial aid beneficiaries also have lower Baccalaureate grades, both when nonadjusted and when adjusted for differences in academic performance between tracks. N = 274,316.

for each student, whether they would have been admitted to a program—independently of whether they had chosen to remain on the waiting list.

Prestige of programs. For each higher education program, we computed the prestige of the program as the average Baccalaureate grade of all the students enrolled in the program. The higher the average Baccalaureate grade, the more prestigious the program.

Prestige gain. For each student, we computed the difference between the prestige of the offer received during Phase 1 and the prestige of the program where the student enrolled. For students who accept the offer during Phase 1, the prestige gain is zero. Students who are admitted in a pending wish can access a more (or less) prestigious program.

Preference gain. Each student had to submit a rank-ordered list of programs where they wanted to apply. We consider the rank of the program as a proxy for preferences, with programs that are ranked higher being preferred to programs

that are ranked lower. Students can only be waitlisted on programs that they ranked higher than the program from which they received an initial offer. This means that by waiting, students receive a final offer from a program they had initially ranked higher. For each student, we computed how much higher the rank of their final offer was compared to the initial offer they had received. For example, a student could get an initial offer from the program in the 10th position. This student could also be waitlisted in a program in the second position. If they were admitted to that program, the rank of their final offer would have increased by $10 - 2 = 8$ ranks.

Academic performance. We measured students' academic performance using two variables: their Baccalaureate grade and their high-school track (the General track, which is further divided into Science, Humanities, and Economic and Social Sciences, the Technological track, and the Professional track). We include high-school track in their academic performance because universities use academic track as a proxy for the level of preparedness of students for higher education programs. Because the content of the Baccalaureate varies according to the student's track, grades are not directly comparable across tracks (58). To take this into account, we adjusted Baccalaureate grades according to baseline (middle school) average academic performance of students enrolled in each track (see [SI Appendix, Table SI.1.3](#) for details on how we adjusted Baccalaureate grades). Differences in academic performance between financial aid beneficiaries and non-financial aid beneficiaries can be found in Table 5.

Benefits of waiting. Students differed in their aspirations, that is the number, type, and prestige of programs they included in their rank-ordered wish list. We recorded the number of pending wishes, i.e., the programs to which students had applied and in which they were waitlisted. We computed the likelihood of being accepted in each pending wish based on the student's ranking on the waitlist and selected the best possible odds across pending wishes. We also recorded the prestige of the offer received during Phase 1 and the potential prestige gains from being accepted in a pending wish. The bigger the potential prestige gains, the more a student benefits from being accepted in a pending wish. Finally, we also recorded the rank of the offer received during the first admission phase. The lower the offer is ranked in the wish list, the more a student can gain by waiting on pending wishes. Differences in aspirations between financial aid beneficiaries and non-financial aid beneficiaries can be found in Table 6.

Cost of waiting. For each student, we computed the distance between the location of their initial offer and the location of the pending program in kilometers. The further away these programs are, the more likely a student will face additional housing costs if they are admitted into a pending program. Some

Table 6. Differences in aspirations according to financial aid status, controlling for academic performance

Variable	Mean non-FA	FA - non-FA controlling for academic performance	P value
Total number of wishes	11.17	0.075	0.03
Applied to selective academic track	0.30	0.005	$P < 0.001$
Applied to selective professional track	0.45	0.009	$P < 0.001$
Applied to selective technical track	0.42	-0.021	$P < 0.001$
Applied to university track (mostly non-selective)	0.84	0.041	$P < 0.001$
Average prestige all wishes	13.07	-0.085	$P < 0.001$
Average prestige all wishes (standardized)	0.58	-0.043	$P < 0.001$
SD average prestige all wishes	1.41	0.011	$P < 0.001$
Average prestige top 3 wishes	13.75	-0.118	$P < 0.001$
SD average prestige top 3 wishes	0.88	0.016	$P < 0.001$
Min average prestige all wishes	10.88	-0.096	$P < 0.001$
Max average prestige all wishes	15.10	-0.054	$P < 0.001$
Number of pending programs	2.02	-0.005	0.53
Chances being accepted in a pending program	-2.18	0.050	0.28
Rank offer phase 1	5.05	0.041	0.02
Prestige offer phase 1	12.69	-0.061	$P < 0.001$
Gain in prestige from pending program	0.37	-0.020	$P < 0.001$

This table shows the difference between non-financial aid beneficiaries (non-FA) and financial aid beneficiaries (FA) in terms of aspirations, controlling for academic performance (high-school track and adjusted Baccalaureate grades). Financial aid beneficiaries apply to less prestigious programs overall. N = 274,316.

higher education programs do not report their location, such that we have 15,484 missing observations for distance between programs (corresponding to 5.6% of our sample). To ensure that this does not affect our analysis, we replicated all our findings on this restricted sample and found similar results, with financial aid beneficiaries being 6% less likely to wait after controlling for academic performance and the benefits of waiting.

Unobserved characteristics. We created dummy variables representing each student's high school to capture unobserved characteristics that may influence their decision to wait, such as neighborhood effects, teacher influences, and peer dynamics. For instance, students' utility functions likely reflect social factors, including the higher utility they derive from maintaining friendships. Since students from lower socioeconomic backgrounds often attend schools with a higher proportion of their peers from similar backgrounds, the peer composition varies between financial aid and non-financial aid students. If students consider their peers' decisions to make their own choices, peer composition differences will affect students' waiting behavior. By incorporating school fixed effects, we control for these peer effects, comparing students within the same high school who share similar peer groups, teachers, and neighborhoods. Teachers may encourage all students, regardless of socioeconomic background, to wait for acceptance into pending programs. Therefore, students in the same school might be more likely to wait based on such advice. By including high school fixed effects in our regression, we isolated the differences in waiting behavior between students of different socioeconomic statuses within the same school, effectively neutralizing the influence of school-related factors.

Analytical Strategy. To address our research question on socioeconomic differences in waiting behavior, we followed several empirical steps. First, we tested whether students receiving financial aid during high school were less likely to wait than non-financial aid beneficiaries when presented with the opportunity to wait. To do this, we estimated a linear probability model with the probability to always wait as the dependent variable and the financial aid status of students as the independent variable. We clustered errors at the high school level to account for similarities between students from the same high school and added year fixed effects to neutralize country-level time-dependent factors that may have led all students of a given cohort to change their waiting behavior relative to the previous cohort.

Second, we investigated whether higher education institutions discriminate against financial aid beneficiaries compared to non-financial aid beneficiaries. To do so, we tested whether socioeconomic status influenced the probability of being admitted into a pending program, after taking into account academic

performance and the prestige of the program. We also investigated whether financial aid beneficiaries and non-financial aid beneficiaries had similar gains of waiting in terms of the increase in the rank of their final offer.

Third, we investigated the role of the costs and benefits of being admitted in a pending program in socioeconomic differences in waiting behavior. To do this, we estimated a linear probability model with the probability to always wait as the dependent variable and the financial aid status plus variables capturing the costs and benefits of waiting as the independent variables. We sequentially added variables linked to the probability of being admitted in a pending program, the value of being admitted in a pending program, and the costs of waiting to investigate the potential role of each of these factors. We interacted the cost of waiting with the financial aid status of students as moving costs may affect students with different socioeconomic statuses differently. We clustered errors at the high school level to account for similarities between students from the same high school and added year fixed effects.

Finally, we added high school fixed effects to our analysis to capture unobserved characteristics that may influence students' decision to wait, such as neighborhood effects, teacher influences, and peer dynamics. Additionally, we ran a separate analysis for each year (2015, 2016, and 2017), as per our preregistration (*SI Appendix, section 4*).

Data, Materials, and Software Availability. Data for this project is hosted by the CASD (Center for Access to Secured Data) at the following address: <https://www.casd.eu/en/source/post-baccalaureate-admission-application-feedback/>. Data are available to researchers upon request. The R scripts associated with this research will be made available to the public at the following address: <https://osf.io/tcfs7/> (59).

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